

A Lean Formalization of Uniform Curve Estimates under Ricci Flow

A module for Perelman’s finite extinction argument

GPT-6 Astra

Juii-hang Leung

September 9, 2026

Abstract

We describe a Lean formalization of length and total-curvature estimates for smooth immersed closed curves evolving by curve shortening in an ambient Ricci flow. The development derives the corrected curvature evolution from actual geometric definitions, treats curvature zeros by regularization, and integrates against the moving arclength measure. For a compact base Ricci flow it constructs finite ambient tensor bounds and proves that the same coefficients apply to products with every supplied static one-dimensional Riemannian metric. The constants are chosen before the factor metrics, curve families, and initial bounds. The result is an interface for the finite extinction argument, conditional on existence of the smooth flows and on uniform initial data. It does not prove flow existence or finite extinction. We explain the corrected curvature estimate, the use of a spatial time-dependent connection, the precise library interfaces, and the scope of the checked artifact.

1 Scope and mathematical setting

Perelman’s finite extinction argument uses controlled curve families in an ambient Ricci flow [3]. The relevant estimates are detailed in Chapter 19 of *Ricci Flow and the Poincaré Conjecture* [4]; the subsequent correction [1] changes the curvature evolution and the dependence on initial data. The formalization reported here concerns this estimate module. It supplies neither the surrounding topological argument nor the analytic existence theory.

All metrics and connections below are genuine smooth Riemannian objects. Curves are represented by maps $c(t, x)$ with $x \in \mathbb{R}$ and period one; integration over $[0, 1]$ therefore represents integration over the parameter circle. An immersion means $c_x(t, x) \neq 0$. Write

$$X = c_x, \quad v = |X|_{g(t)}, \quad S = v^{-1}X, \quad H = D_s S, \quad q = |H|_{g(t)}^2, \quad k = \sqrt{q},$$

where $D_s = v^{-1}D_x$ is the spatial Levi-Civita derivative along the curve. The curve-shortening equation is $c_t = H$, while the ambient equation is $\partial_t g = -2\text{Ric}_g$. The geometric quantities controlled are

$$L(t) = \int_0^1 v(t, x) dx, \quad \Theta(t) = \int_0^1 k(t, x)v(t, x) dx, \quad E(t) = \int_0^1 q(t, x)v(t, x) dx.$$

These definitions use the evolving metric, rather than the fixed norm of a chart model. No global chart containing the curve is assumed; self-intersections are allowed.

The finite-dimensional manifolds are smooth, boundaryless, Hausdorff, and sigma-compact; their models are real inner-product spaces of positive dimension. The principal ambient manifold is compact. Precise Lean typeclass assumptions, including the time-set convention, are recorded in

Appendix A. The metrics are jointly smooth on the specified closed time interval in the library’s within-smoothness sense. The supplied curve flows are jointly smooth, closed, immersed, and satisfy the actual within-derivative flow equation. These conditions express an existing solution, not a promise to construct one.

2 Uniform statement and the order of quantifiers

Theorem 2.1 (Uniform products with a static one-dimensional factor). *Let M be a compact manifold with the hypotheses above, and let $g(t)$ be a smooth Ricci flow on a nondegenerate closed interval $[a, b]$. Let N be a supplied smooth boundaryless Hausdorff sigma-compact one-dimensional manifold. There exist $B, C \geq 0$ with the following property. For every pair of index types $\Lambda, \mathcal{I}$, every family of smooth Riemannian metrics h_ℓ on N , and every family of smooth immersed closed curve-shortening flows $c_{\ell,i}$ in $(M \times N, g(t) \oplus h_\ell)$, the following holds. For every $L_0, \Theta_0 \in \mathbb{R}$, if*

$$\forall \ell, i, \quad L_{\ell,i}(a) \leq L_0, \quad \Theta_{\ell,i}(a) \leq \Theta_0,$$

then, for all ℓ, i and all $t \in [a, b]$,

$$L_{\ell,i}(t) \leq L_0 e^{B(t-a)}, \quad \Theta_{\ell,i}(t) \leq (\Theta_0 + L_0) e^{C(t-a)}. \quad (1)$$

Each h_ℓ is independent of Ricci-flow time. No regularity in either index, and no compactness of N , is required.

Lean declaration. `CurveControl.Geometry.exists_uniform_static_product_family_estimate`
Module. `CurveControl/Geometry/UniformProductEstimates.lean`

The constants are selected before $\Lambda, \mathcal{I}, h, c, L_0, \Theta_0$. The proof obtains them from the compact base metric family and the time interval; it does not inspect the factor metric or the curves when choosing witnesses. The fixed factor manifold and model are parameters of the Lean context. The witness construction itself uses only the base. Initial bounds remain explicit inputs; they are not extracted from a compact parameter family by this module.

Corollary 2.2 (Positive-scale families). *Theorem 2.1 holds with $\Lambda = (0, \infty)$ and arbitrary supplied static metrics h_λ . The coefficients do not require a positive lower bound on λ , nor continuity or differentiability of h_λ in λ .*

Lean declaration. `CurveControl.Geometry.exists_uniform_positive_scale_product_estimate`
Module. `CurveControl/Geometry/UniformProductEstimates.lean`

The supplied one-dimensional manifold may in particular be a smooth circle. The example module constructs the genuine metric $h_\lambda = \lambda^2 h_*$ from any supplied smooth Riemannian metric h_* , proves $|V|_{h_\lambda} = \lambda |V|_{h_*}$, and instantiates the uniform theorem for all positive scales. A particular `AddCircle` atlas and its starting metric remain caller inputs. The same example gives bounds uniform over the entire time interval, with right sides $L_0 e^{B(b-a)}$ and $(\Theta_0 + L_0) e^{C(b-a)}$.

Lean declaration. `CurveControl.Examples.StaticFactorApplication.scaledFactorMetric`
Module. `CurveControl/Examples/StaticFactorApplication.lean`

Lean declaration. `CurveControl.Examples.StaticFactorApplication.exists_scaledFactor_fullInterval_bounds`

Module. `CurveControl/Examples/StaticFactorApplication.lean`

The proof of Theorem 2.1 is the combination of the following sections.

3 The corrected local evolution

The correction to Section 19.2 [1] identifies the omitted time variation of the ambient connection. It produces an error linear in k , as well as quadratic terms, and consequently a total-curvature estimate involving initial length. The implementation retains this term but uses a spatial formulation instead of constructing the full space-time metric.

Fix a valid spatial chart near the point of computation. For a vector field along c , use

$$D_t W = \partial_t W + \Gamma_{g(t)}(H, W), \quad D_x W = \partial_x W + \Gamma_{g(t)}(X, W).$$

In particular D_t is not the metric-compatible derivative of $g(t) + dt^2$. Its product rule contains the ambient metric variation:

$$\partial_t \langle V, W \rangle = -2 \text{Ric}(V, W) + \langle D_t V, W \rangle + \langle V, D_t W \rangle. \quad (2)$$

Let $A = \partial_t \nabla^{g(t)}$, with the spatial coordinate held fixed. The actual connection variation theorem gives

$$\langle A(U, V), W \rangle = -(\nabla_U \text{Ric})(V, W) - (\nabla_V \text{Ric})(U, W) + (\nabla_W \text{Ric})(U, V). \quad (3)$$

This is the lowering of $\partial_t \Gamma$, not the derivative of a lowered Christoffel symbol. Equation (2) supplies the separate derivative of the metric.

For signs, let $R^+(U, V)W = D_U D_V W - D_V D_U W - D_{[U, V]} W$. The upstream do Carmo convention is $R^- = -R^+$. In the evolution formula write

$$\text{rm}(U, V, W, Z) = \langle R^-(U, V)W, Z \rangle.$$

The last-slot antisymmetry yields $\text{rm}(H, S, H, S) = \langle R^+(H, S)S, H \rangle$. The coordinate-to-intrinsic bridge proves this equality with its signs; it does not identify the two curvature operators.

Put $\alpha = q + \text{Ric}(S, S)$ and $P = D_s H + qS$. The velocity and connection commutators give

$$v_t = -\alpha v, \quad D_t S = D_s H + \alpha S, \quad D_t H = D_s^2 H + 2\alpha H + (D_s \alpha)S + R^+(H, S)S + A(S, S).$$

Since $\langle H, S \rangle = 0$ and $\langle D_s H, S \rangle = -q$, P is the spatial normal component of $D_s H$, and $|D_s H|^2 = |P|^2 + q^2$.

Proposition 3.1 (Actual squared-curvature evolution). *For an existing smooth immersed closed curve-shortening flow in a smooth ambient Ricci flow on a time interval J , at each $t \in \text{interior } J$ and each parameter x , one has*

$$\begin{aligned} q_t = & D_s^2 q - 2|P|^2 + 2q^2 + 4q \text{Ric}(S, S) - 2 \text{Ric}(H, H) \\ & + 2 \text{rm}(H, S, H, S) - 4(\nabla_S \text{Ric})(S, H) + 2(\nabla_H \text{Ric})(S, S). \end{aligned} \quad (4)$$

Lean declaration. `CurveControl.Geometry.IsCurveShorteningFlowOn.hasDerivAt_curvatureSq_time`
Module. `CurveControl/Geometry/CurvatureEvolution.lean`

The hypotheses of this declaration include the actual two flows, the geometric context in Appendix A, an interior time, and a parameter. They do not include a proposed curvature PDE, a Kato inequality, a connection variation identity, or differentiability of H as additional assumptions. The latter facts are derived using fixed-chart neighborhoods and equality of function germs. Choosing the chart centered at the current point occurs only after the differentiated identities have been established.

The relation to the space-time projection can be checked explicitly. If $T = \partial_t$ and $\beta = \text{Ric}(S, H)$, that projection is $\hat{P} = P + \beta T$. Its squared norm contains β^2 , absent from the spatial P . The corresponding space-time curvature terms cancel this extra square and the mixed quadratic Ricci term; the remaining connection derivatives are precisely the last two terms of (4). This explains why copying a space-time projection into a spatial formula would be incorrect.

4 Regularization and the moving integral

Let $B, K, D \geq 0$ bound the ambient Ricci tensor, curvature tensor and covariant derivative of Ricci, respectively, in the evolving metric. A bound is an estimate on evaluation on arbitrary vectors, by the product of their metric norms; it is not a bound on unrelated chart coordinates. Because $|S| = 1$ and $|H| = k$, the last five terms of (4) have absolute value at most

$$(6B + 2K)q + 6D\sqrt{q} \leq A_0(q + \sqrt{q}), \quad A_0 = \max(6B + 2K, 6D).$$

The geometric Kato estimate is $(D_s q)^2 \leq 4q|P|^2$. For $\varepsilon > 0$, define $h_\varepsilon = \sqrt{q + \varepsilon^2}$. Differentiation, the Kato estimate, and the nonpositive dissipation term give

$$(h_\varepsilon)_t \leq D_s^2 h_\varepsilon + k^3 + \frac{A_0}{2}(h_\varepsilon + 1). \quad (5)$$

There is no differentiation of k at a zero of q .

Lean declaration. `CurveControl.Analysis.deriv_regularizedNorm_le`

Module. `CurveControl/Analysis/RegularizedPDE.lean`

Let $r = \text{Ric}(S, S)$ and $\Theta_\varepsilon = \int_0^1 h_\varepsilon v \, dx$. The true density equation is $v_t = -(q + r)v$. Inserting it in the differentiated integral, using periodicity to eliminate the diffusion flux, and using $k^3 - h_\varepsilon q \leq 0$ yields

$$L' \leq BL - E, \quad \Theta'_\varepsilon \leq (A_0/2 + B)\Theta_\varepsilon + (A_0/2)L.$$

Thus $C = A_0/2 + B$ controls $\Theta_\varepsilon + L$. Parameter differentiation, continuity, integrability, and the periodic endpoint equality are constructed from the actual smooth functions. The geometric caller does not supply them.

Lean declaration. `CurveControl.Analysis.ScalarCurveHypotheses.regularized_comparison`

Module. `CurveControl/Analysis/CurveEstimateAssembly.lean`

The comparison argument uses derivatives only at interior times and continuity on the closed time interval. In particular it does not claim a two-sided time derivative at the initial endpoint. For each fixed time,

$$0 \leq \Theta_\varepsilon - \Theta \leq \varepsilon L.$$

Taking limits in the already established comparison, at both its initial and final times, avoids exchanging a limit with a time derivative.

Lean declaration. `CurveControl.regularizedWeightedCurvature_sub_bounds`

Module. `CurveControl/Analysis/RegularizedIntegral.lean`

Theorem 4.1 (Geometric estimate with explicit ambient bounds). *Let g, c be the actual flows described above on J . Let $a \leq b$ and $[a, b] \subseteq J$, and suppose the actual nonnegative ambient tensor bounds B, K, D hold at every point and time of J . With $C = \max(6B + 2K, 6D)/2 + B$, for all $t \in [a, b]$,*

$$L(t) \leq L(a)e^{B(t-a)}, \quad \Theta(t) + L(t) \leq (\Theta(a) + L(a))e^{C(t-a)}.$$

Lean declaration. `CurveControl.Geometry.IsCurveShorteningFlowOn.curve_estimate`

Module. `CurveControl/Geometry/CurveEstimates.lean`

The intermediate structure `CurveControl.Analysis.ScalarCurveHypotheses` is an explicit conditional analysis interface containing a squared-curvature PDE, Kato, and density assumptions. The geometric theorem proves its fields from the same g, c ; passing that structure alone is not the geometric result.

Proposition 4.2 (Accumulated curvature energy). *Under the same flow and interval hypotheses, it is enough to assume $\text{Ric}(V, V) \geq -B$ for every unit vector $(g(t)(V, V) = 1)$, at every point and time of $[a, b]$, to obtain*

$$\int_a^b e^{-B(t-a)} E(t) dt \leq L(a) - e^{-B(b-a)} L(b).$$

If also $B \geq 0$, then $\int_a^b E(t) dt \leq e^{B(b-a)} L(a) - L(b)$.

Lean declaration. `CurveControl.Geometry.IsCurveShorteningFlowOn.weighted_curvatureEnergy_integral_le`

Module. `CurveControl/Geometry/LengthEstimates.lean`

Lean declaration. `CurveControl.Geometry.IsCurveShorteningFlowOn.curvatureEnergy_integral_le`

Module. `CurveControl/Geometry/LengthEstimates.lean`

The blueprint’s integral-evolution node additionally states local absolute continuity of unregularized Θ and an almost-everywhere differential inequality. Those statements have not been implemented literally here. The regularized comparison followed by a limit is a sufficient alternative for Theorem 4.1. A downstream argument requiring that AC/a.e. interface would need a further theorem.

5 Ambient constants from compactness

Proposition 5.1 (Finite actual tensor bounds). *Let M be compact with the geometric context of Appendix A, and let g be a jointly smooth metric family on a time set J . For $[a, b] \subseteq J$ there exist $B, K, D \geq 0$ bounding the actual Ric , rm and ∇Ric everywhere on $M \times [a, b]$, in the multilinear metric-norm sense used above. No Ricci-flow equation is needed for this proposition.*

Lean declaration. `CurveControl.Geometry.CompactAmbientBounds.exists_uniformTensorBoundsOn`

Module. `CurveControl/Geometry/CompactAmbientBounds.lean`

The proof passes through finitely many compact chart pieces. Joint smoothness provides continuity of the relevant tensor coefficients and the inverse Gram matrix. On each compact piece, finite coefficient families are bounded. Metric norm control of the coordinates converts these coefficient bounds into intrinsic multilinear estimates. A finite cover combines the local nonnegative constants into global ones. This step avoids assuming an unproved globally continuous operator-norm bundle section. All coefficients in the construction are those of the canonical geometric tensors; no surrogate curvature field is supplied.

Applying this proposition to the base Ricci flow fixes B, K, D independently of every later auxiliary metric. What remains is to preserve the same constants when passing to a product, without dependence on the geometry of the small factor. Compactness of the product, or a uniform lower injectivity-radius bound, is not used for this purpose.

6 The genuine product and one-dimensional uniformity

The product construction starts with the upstream metric `DCProductMetric`. A technical distinction matters: the ordinary norm on $E \times F$ is a maximum norm, whereas the canonical intrinsic interfaces require an inner-product model. The development uses the continuous linear equivalence to the L^2 type `WithLp2` on $E \times F$ and transports the model with corners. It constructs the induced genuine metric on the same product manifold.

This is not an assertion that the maximum norm is Hilbertian. Extended charts, tangent-coordinate changes, and the chart Gram pairing are transported explicitly. Differentiating the Gram identity and applying the algebraic uniqueness of a symmetric metric-compatible connection proves the actual Christoffel formula. Differentiating that formula on an open chart target then proves the curvature formula.

Lean declaration. `CurveControl.Geometry.chartChristoffelContraction_productL2`

Module. `CurveControl/Geometry/ProductL2Connection.lean`

Lean declaration. `CurveControl.Geometry.chartCurvature_productL2`

Module. `CurveControl/Geometry/ProductL2Connection.lean`

For intrinsic curvature, an isometry is built from the actual tangent-fiber metrics, not from model-space norms. The four-tensor splits into the sum of the two factor tensors. Ricci splitting follows by tracing in orthonormal bases under this actual isometry. Finally the covariant Ricci derivative is identified with $\partial \text{Ric} - \Gamma \text{Ric} - \Gamma \text{Ric}$ in coordinates; the derivative of the lower-order splitting is justified on a neighborhood.

Lean declaration. `CurveControl.Geometry.canonicalCurvatureFormAt_productL2`

Module. `CurveControl/Geometry/ProductIntrinsicCurvature.lean`

Lean declaration. `CurveControl.Geometry.ricciTensorAt_productL2`

Module. `CurveControl/Geometry/ProductIntrinsicCurvature.lean`

Lean declaration. `CurveControl.Geometry.ProductCovRicci.covRic_productL2`

Module. `CurveControl/Geometry/ProductCovRicci.lean`

Proposition 6.1 (Preservation of the base constants). *Let g and h be genuine smooth metrics on the manifolds M, N above, with $\dim N = 1$. If B, K, D bound the three actual base tensors at $p \in M$, the same constants bound the product tensors at every (p, z) . For a base Ricci flow and static h , $g(t) \oplus h$ is itself a Ricci flow on the same time interval.*

Lean declaration. `CurveControl.Geometry.productL2_tensorBounds_of_finrank_one`

Module. `CurveControl/Geometry/UniformProductEstimates.lean`

Lean declaration. `CurveControl.Geometry.isRicciFlowOn_productL2_oneDimensional`

Module. `CurveControl/Geometry/ProductFlow.lean`

Indeed, antisymmetry of the algebraic curvature tensor and collinearity of all vectors in dimension one give zero canonical curvature. Consequently the actual Ricci field vanishes identically; differentiating this field gives zero covariant Ricci. It would not suffice to infer a derivative from vanishing at a single point. The second-factor terms in all product splittings therefore vanish. Each projection decreases the actual metric norm, so the base constants are unchanged.

Lean declaration. `CurveControl.Geometry.canonicalCurvatureFormAt_eq_zero_of_finrank_one`

Module. `CurveControl/Geometry/OneDimensionalCurvature.lean`

Lean declaration. `CurveControl.Geometry.covRicciAt_eq_zero_of_finrank_one`

Module. `CurveControl/Geometry/OneDimensionalCurvature.lean`

For the time-dependent statement, joint within-smoothness of the product metric is proved by pulling back the base family and the static factor form along the two projections and adding them. The within-time derivative has no factor contribution. Actual Ricci splitting identifies the resulting derivative with -2Ric of the product. Thus product flow status is a conclusion. Combining Propositions 5.1 and 6.1 with Theorem 4.1 proves Theorem 2.1; the initial bound for $\Theta + L$ is $\Theta_0 + L_0$, and $L(t) \geq 0$ yields (1).

7 Implementation, provenance, and limits

The dependency base is mathlib at commit 520045ab14e26149ee970e2e617ca04b09bde5d6 and the frenzymath source tree at commit bb91a091f0b968f8bbe8d861e025a88d82b161be [5, 2]. The project imports the upstream Ricci-flow and Riemannian geometry libraries. These libraries provide the smooth Riemannian metric, canonical Levi–Civita connection, chart tensor, and Ricci-flow infrastructure. mathlib provides the general manifold, calculus, linear algebra, and integration framework. The newly supplied development consists of the curve-specific definitions and bridges, spatial evolution proof, regularized integral chain, compact tensor bounds, product transport and splitting, and final quantified integration of these parts. We make no claim of a new mathematical estimate or of priority for its formalization.

The frozen blueprint’s labels `thm:curve-flow-curvature-evolution`, `lem:regularized-curve-curvature-inequality`, and `thm:curve-flow-length-total-curvature-bounds` describe the corresponding mathematical goals. The time-space connection and AC/a.e. intermediate nodes are handled by the alternatives explained above; a matching final conclusion should not be confused with literal implementation of every blueprint node.

The release record reports a successful complete target build with Lean 4.32.1 and an environment-level reachable-axiom check of the project declarations. The reachable axioms were only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx` or additional axiom was found. Dependency fingerprints and source hashes are bound to the release record. The build reused dependency caches, so it is not a claim of reproduction on a clean machine. The machine check establishes correctness relative to the definitions and axioms; matching the mathematical specification also requires semantic review.

AI assistance was used in development. Separate Astra agents reviewed proofs they had not authored; these reviews were not external human peer review.

The principal limitations are explicit inputs and omitted neighboring modules: existence and continuation of the smooth Ricci and curve flows; construction of concrete circle atlases and their starting metrics; construction of ramps and their uniform initial bounds; and the remaining geometric and topological steps of finite extinction. No acceptance by an external mathematical team is asserted. The delivered interface can be used once those objects and hypotheses have been supplied.

A Exact interface assumptions

Project declaration names in this report are fully qualified; the upstream Ricci-flow predicate is written as `IsRicciFlowOn`. The main product roots use the following context; symbols are printed in ASCII to keep the inventory independent of a Lean font setup.

```
E, F : Type*; H, K, M, N : Type*
NormedAddCommGroup E; InnerProductSpace Real E
NormedAddCommGroup F; InnerProductSpace Real F
FiniteDimensional Real E; FiniteDimensional Real F
NeZero (Module.finrank Real E)
NeZero (Module.finrank Real F)
TopologicalSpace H; TopologicalSpace K
I : ModelWithCorners Real E H
J : ModelWithCorners Real F K
I.Boundaryless; J.Boundaryless
TopologicalSpace M; ChartedSpace H M; IsManifold I infinity M
```

```

TopologicalSpace N; ChartedSpace K N; IsManifold J infinity N
SigmaCompactSpace M; T2Space M
SigmaCompactSpace N; T2Space N; CompactSpace M

```

The explicit arguments are a metric family g of type `Real→Riemannian.RiemannianMetricIM`, $a, b : \mathbb{R}$, `IsRicciFlowOn(Set.Iccab)`, $a \leq b$, and `Module.finrankRealF=1`. In the inventory, `Real` and `infinity` stand for Lean’s real type and smoothness order ∞ . The base estimate uses only the E, H, I, M part of the context, without `CompactSpaceM`, and takes actual uniform tensor bounds as input. The compact-bound theorem also lists `CompleteSpaceE`; this instance is available from finite dimensionality over the complete real field, and is inferred when the final product root calls it.

`IsRicciFlowOn` has four fields: order-connectedness and nontriviality of the time set, joint smoothness of the metric as a horizontal bilinear-form section, and the within-time equation evaluated on every point and pair of tangent vectors. Hence its specialization to $[a, b]$ rules out a singleton interval even though the root also lists $a \leq b$.

`CurveControl.Geometry.IsCurveShorteningFlowOn` requires joint `ContMDiffOn` smoothness on the time set times $\mathbb{R}$, period one at each time in that set, nonzero actual velocity at every such time and parameter, and a `HasDerivWithinAt` equation for the local time curve whose derivative is the actual curvature vector. These are the complete substantive flow conditions; no target inequality is a field.

The final quantifier order, suppressing only this fixed context, is

$$\begin{aligned}
& \exists B, C \geq 0 \quad \forall \Lambda, \mathcal{I}, h, c, \\
& \text{(actual curve flows)} \implies \forall L_0, \Theta_0, \\
& \text{(initial bounds)} \implies \forall \ell \in \Lambda \quad \forall i \in \mathcal{I} \quad \forall t \in [a, b] \quad (1).
\end{aligned}$$

The types $\Lambda, \mathcal{I}$ are arbitrary; an empty family is allowed. The positive-scale root fixes Λ to the subtype $\{\lambda : \mathbb{R} \mid 0 < \lambda\}$ before quantifying over the supplied metric family. The factor metric has no time argument. Neither a curvature-splitting hypothesis nor a product Ricci-flow hypothesis appears in either final root.

References

- [1] Correction to Section 19.2 of Ricci Flow and the Poincaré Conjecture. arXiv:1512.00699, 2015. Version 1, December 2, 2015. <https://arxiv.org/abs/1512.00699>.
- [2] frenzymath contributors. Poincare-Conjecture: blueprint and formalized sources. Source repository. Source snapshot bb91a091f0b968f8bbe8d861e025a88d82b161be; includes Riemannian geometry and Ricci-flow libraries.
- [3] Grisha Perelman. Finite extinction time for the solutions to the Ricci flow on certain three-manifolds. *arXiv preprint math/0307245*, 2003. Submitted July 17, 2003.
- [4] Ricci flow and the Poincaré conjecture. Clay Mathematics Monographs, volume 3. American Mathematical Society and Clay Mathematics Institute, 2007. <https://www.claymath.org/library/monographs/cmim03.pdf>.
- [5] The mathlib Community. mathlib4: mathematics in Lean. Source repository. Dependency pinned to commit 520045ab14e26149ee970e2e617ca04b09bde5d6.